Yash Kokane

Dual Major Undergraduate at IIT Gandhinagar
Materials Science Engineering and Computer Science Engineering

yash.kokane@iitgn.ac.in
+91 9321697613

Degree	Institution	CPI/%	Year
Dual Major B.Tech	IIT Gandhinagar	8.06	2020 – Present
Senior Secondary	Pace Junior Science College	84.31%	2020
Secondary	A.K. Joshi School	94.4%	2018

Publications

- **Active-Learning Assisted General Framework for Efficient Parameterization of Force-Fields**
  Yati*, Kokane, Y.; Mondal, A.
  Manuscript accepted at the Journal of Chemical Theory and Computation. DOI: 10.1021/acs.jctc.5c00061
- **Capacitive Performance of Palladium-activated Charcoal as an Electrode Material**
  Khan, F. J.; Mane, A.; Mali, R.; Kokane, Y.; Kokane, S.*
  Manuscript under review at Current Physical Chemistry
- **Toward sustainable polymer design: a molecular dynamics-informed machine learning approach for vitrimers**
  Zheng, Y.; Kokane, Y.; Biswal, A.; Guo, Y.; Vashisth, A.*
  Manuscript accepted at in Digital Discovery. DOI: 10.1039/D5DD00239G

Achievements and Honors

- Winner - ML Problem Statement, HackRush 2024 (Hackathon conducted by IIT Gandhinagar).
- Recipient of the Dr. T. G. Visweswaraiah Scholarship, IIT Gandhinagar (2024).
- First Prize - ES 301 Mechanics of Solids course presentation on analysing FDM printing infill parameters (2022).
- Secured Middle School and High School Maharashtra State Government Scholarships through state-level examinations.
- Represented India at the 32$^{\text{nd}}$ China Adolescent Science and Technology Innovation Contest (CASTIC) 2017, won the bronze medal among 25+ participating countries.

Internships

- **Machine Learning for prediction and screening of vitrimers with desired properties** (May – July 2024)
  Summer Research Internship, University of Washington *(Presentation)*
  - Under the guidance of Prof. Aniruddh Vashisth, trained and benchmarked multiple machine learning algorithms in conjunction with molecular representations such as molecular fingerprints and descriptors to predict the glass transition temperature (Tg) of vitrimers, a novel class of self-healing polymers.
  - Screened over one million vitrimer compositions using ML models trained on 2000+ vitrimer $T_g s$ using high throughput MD to identify those with target T_g values.
  - Utilized SHAP analysis to pinpoint substructures significantly influencing glass transition behavior.
- **Optimization and MPC Modelling for Acoustic Wave Separators** (May – July 2023)
  Summer Internship, TCS Research *(Report)*
  - Under the guidance of Dr. Venkataramana Runkana, developed a mechanistic model for Acoustic Wave Separators (AWS), an advanced biopharmaceutical device used in cell culture clarification. Optimized AWS control parameters using Genetic Algorithms to enhance product throughput.
  - Formulated a Model Predictive Control (MPC) framework to dynamically adjust control parameters, considering scalability for industrial applications.
- **Invention Factory — IIT Gandhinagar** (May – July 2022)
  Design and Prototyping Programme conducted by the Cooper Union, New York *(Provisional Patent Application)*
  - Collaborated with 40+ professors and entrepreneurs from the Indian Institutes of Technology (IITs) to conceive novel mechanical and robotic solutions that can positively impact society.
  - Patented *"LiftSafe"* (Provisional Patent Application Number: 202221042912), a novel portable weightlifting safety device designed to prevent weights from endangering users. Developed functional prototypes, and pitched the product to potential investors.
- **Speech-to-text Model for Marathi Language using deep learning** (Aug – Nov 2023)
  Industrial Internship at PinnacleWorks Infotech *(Internship Completion Certificate)*
  - Developed Acoustic Models for Marathi language speech-to-text conversion using the Kaldi Deep Learning toolkit. Scraped and processed 10M+ Marathi sentences from Wikipedia and news sources for language model training.
  - Tuned Deep Neural Network hyperparameters to develop a working Marathi speech-to-text model with an industry-level Word Error Rate of only 9%.

Projects

- **Optimization of Lennard-Jones parameters for non-polarizable force fields** (Jan – Sept 2024)
  Research Advisor: Prof. Anirban Mondal *(Github Link)*
  - Under the guidance of Prof. Anirban Mondal, developed a Python-based active learning framework integrated with GROMACS, utilizing Gaussian Process Regression coupled with Genetic Algorithms for determining optimal Lennard-Jones parameters for OPLS force fields in Molecular Dynamics Simulations.
  - The optimized LJ parameters accurately modeled density and structural properties for sulfone-based molecules, outperforming published OPLS LJ parameters.
  - Manuscript accepted at the *Journal of Chemical Theory and Computation.* DOI: 10.1021/acs.jctc.5c00061

- **Benchmarking Machine Learning Potentials for High Entropy Alloys** (Aug 2024 – Present)
  Research Advisor: Prof. Raghavan Ranganathan
  - Benchmarked Machine Learning Interatomic Potentials (MLIPs), including DeePMD, SNAP, MACE, and NeQUIP, trained on ab-initio Molecular Dynamics data of High Entropy Alloys (Fe, Mn, Co, Cr, Ni) generated using VASP, featuring defect structures such as twinning, stacking faults, and dislocations.
  - Evaluated potentials on training time, energy and force accuracy, and generalizability to diverse defect configurations using LAMMPS.
- **Experimental Analysis of Stokes-Einstein & Stokes-Einstein-Debye Equations** (Aug 2022 – Nov 2022)
  BS 191 Matter and Energy Laboratory, IIT Gandhinagar *(Report)*
  - Synthesized monodisperse polystyrene colloidal particles via a two-step process, involving initial seed latex preparation followed by controlled seeded growth to achieve uniform particle size.
  - Ensured purity and stability of the colloidal suspension through dialysis to remove impurities.
  - Calculated the diffusion constant of particles by tracking their motion under a digital optical microscope and using OpenCV image analysis, applying the Stokes-Einstein and Langevin equations.
- **Investigating Infill Parameters Involved in 3D Printing** (April 2022)
  Course Project: Mechanics of Solids (ES 221) *(Report)*
  - Led a team of five members for an in-depth investigation into the influence of infill patterns and material parameters on the mechanical properties of FDM 3D-printed components.
  - Conducted mechanical tests using a Universal Testing Machine to evaluate tensile strength and failure modes of specimens with varying infill patterns and densities.
  - Validated experimental results through stress analysis simulations in Autodesk Fusion 360.
- **Identifying Crack-Inducing Parameters in the Continuous Casting Process of Steel** (Jan 2023)
  Research Advisor: Prof. Amit Arora
  - Worked on an industrial project offered by ArcelorMittal to identify crack-inducing parameters in the Continuous Casting process of steel at the Hazira steel production plant.
  - Analyzed time series data from various plant sensors, used feature importances of Random Forest Classifiers to pinpoint influential factors.
  - Designed and implemented a Recurrent Neural Network (RNN) to predict crack propagation in steel slabs.

Teaching Experience

- **Teaching Assistant, MOOC Course on ML assisted Material Discovery** (December 2023)
  - Teaching Assistant for the MOOC Course conducted by IIT Kharagpur, taught by Prof. Subramanian Sankaranarayanan (University of Illinois Chicago) and Prof. Raghavan Ranganathan (IIT Gandhinagar).
- **Academic Discussion Hours Mentor, IIT Gandhinagar** (April 2023 – July 2023)
  - Academic guide for conducting out-of-classroom lectures for academically struggling undergraduate students, selected based on excellent academic performance.
  - Academic mentor for two courses: *Introduction to Materials Science* and *Materials of the Future.*

Positions of Responsiblity

- **Class Representative, Materials Science Department** (August 2022 – August 2024)
  - Acted as a bridge between classmates and faculty to address academic and administrative matters.
  - Effectively communicated class feedback, fostering collaborative improvements.
- **Student Senator, IIT Gandhinagar** (March 2023 – March 2024)
  - Responsible for voicing student concerns to the administration and authorities, conducting surveys and ensuring the overall well-being of students at IIT Gandhinagar.
  - Led the documentation committee, maintaining records of senate proceedings and student constitution revisions.
- **Core member, Robotics Club, IIT Gandhinagar** (May 2021 – Aug 2022)
  - Ideated and conducted multiple workshops on Robotics and Arduino Interfacing for the student community of IIT Gandhinagar.
  - Managed inventory and contributed to long-term robotics projects of the Club.

Relevant Courses

Computational Materials (only student to receive A+ or 11/10 for excellence), Advanced Materials (A or 10/10), Nanoscale Science (A or 10/10), Metallurgy and Extraction (A or 10/10), Introduction to Materials, Physics of Materials, Corrosion and Degradation of Materials, Material Characterization Techniques, Machine Learning, Introduction to Data Science, Deep Learning Specialization courses by Andrew Ng on Coursera. (LinkedIn)

Technical Skills Summary

- **Chemistry Software Packages**: CP2K, GROMACS, VASP, LAMMPS, OVITO.
- **Programming Languages**: Python, Matlab, C, C++, Assembly.
- **Platforms & Frameworks**: PyTorch, Scikit-learn, GitHub, Linux, High-Performance Computing, Microcontrollers (Arduino, NodeMCU).